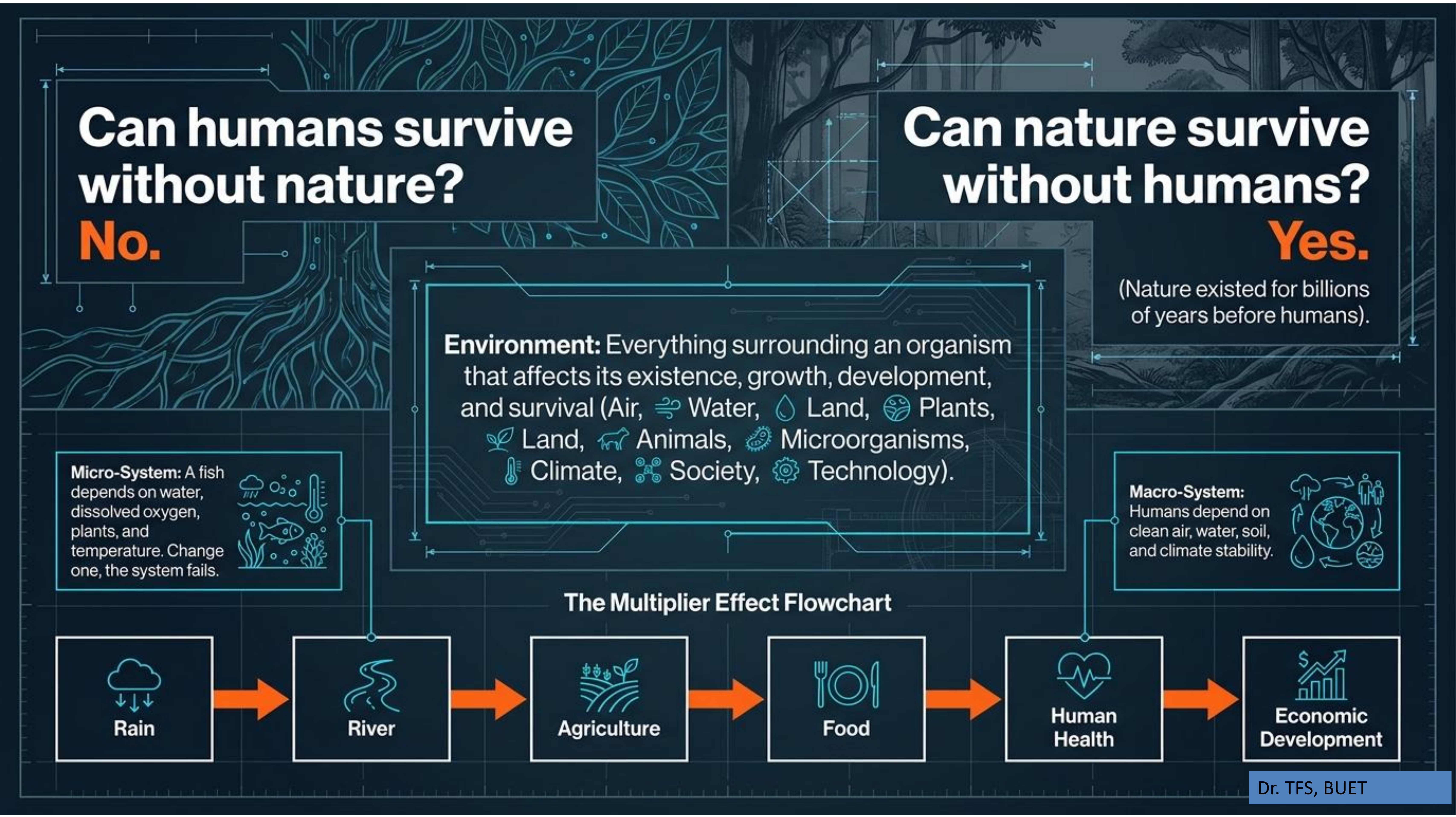
Foundations of Environment and Sustainability

GESL-3211 Environment, Sustainability and Law

Dr. TFS, BUET

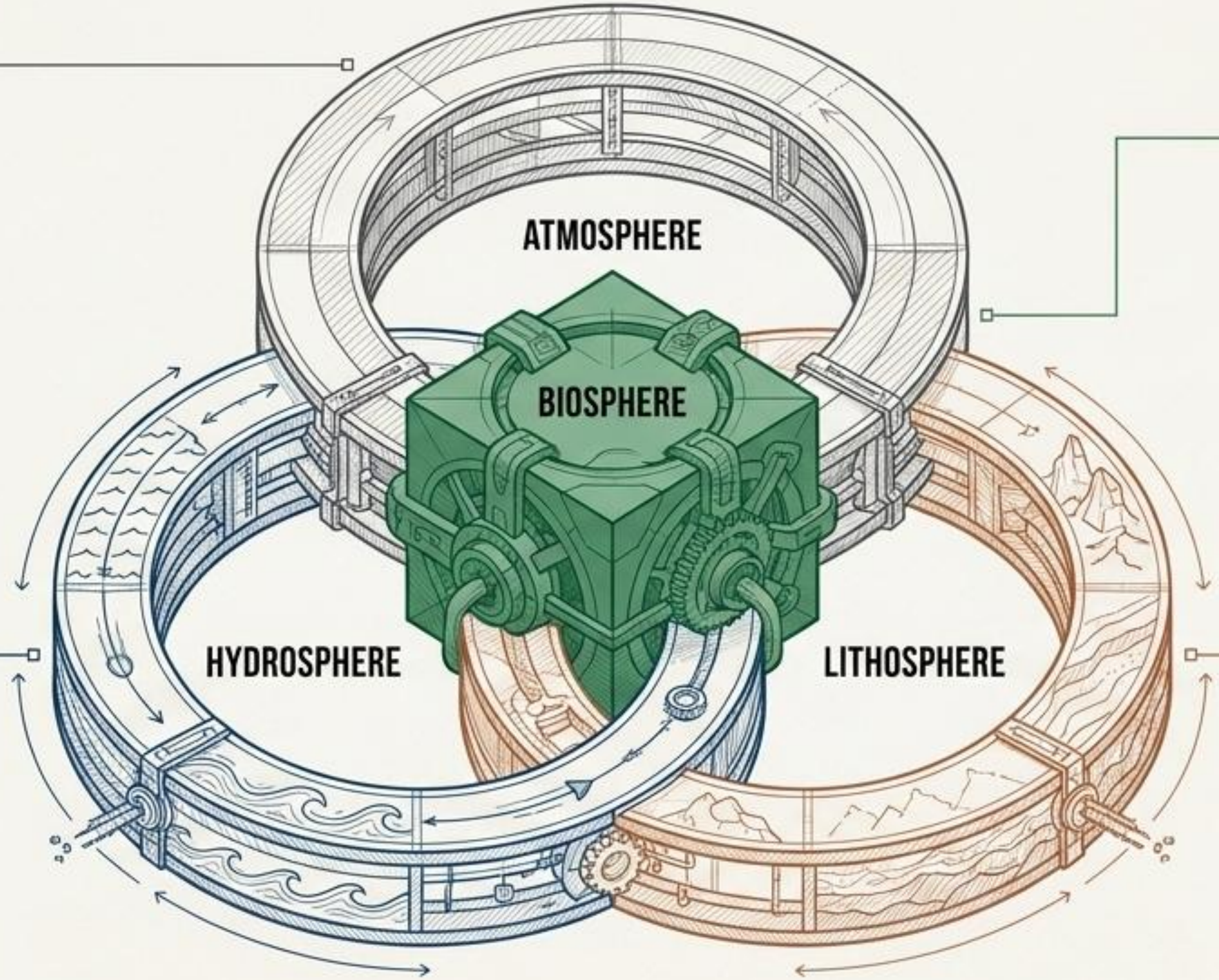

Can humans survive without nature?
No.
Can nature survive without humans?
Yes.
(Nature existed for billions of years before humans).
Environment: Everything surrounding an organism that affects its existence, growth, development, and survival (Air, Water, Land, Plants, Land, Animals, Microorganisms, Climate, Society, Technology).
Micro-System: A fish depends on water, dissolved oxygen, plants, and temperature. Change one, the system fails.
Macro-System: Humans depend on clean air, water, soil, and climate stability.
The Multiplier Effect Flowchart
Rain
River
Agriculture
Food
Human Health
Economic Development

Earth's Physical Operating Systems

FOUR INTERACTING SPHERES FORM THE FOUNDATIONAL MECHANICS OF THE EARTH

Atmosphere:
The gaseous protective layer.

Biosphere:
The exclusive zone where life exists, fundamentally dependent on continuous inputs from the other three spheres.



Hydrosphere:
The total water inventory (surface, ground, vapor, ice).

Lithosphere:
Earth's solid outer crust (rocks, minerals, soil).

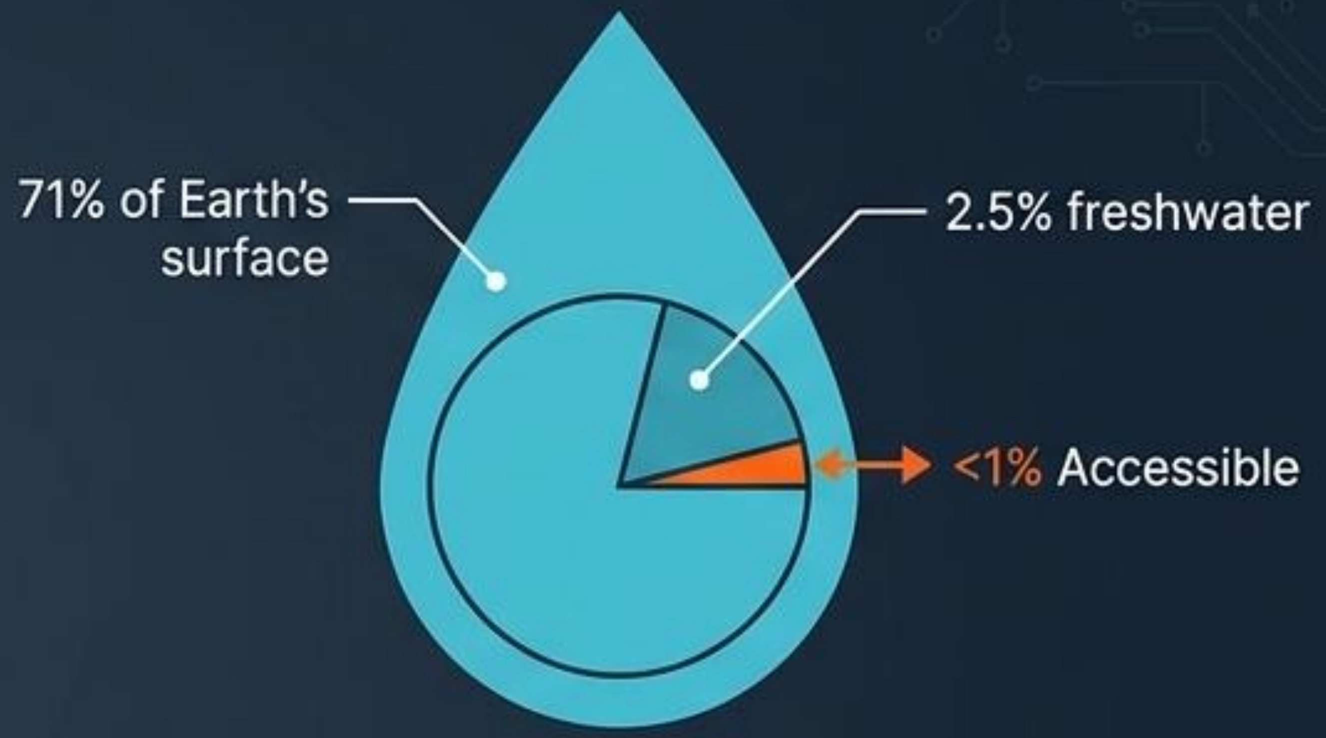
Atmosphere

Plus: Water vapor, Methane, Ozone.

Functions: Provides O2, blocks UV, controls weather.

Stress Test (Bangladesh): Dhaka's poor air quality driven by brick kilns, traffic emissions, and construction dust. Linked to engineering challenges: carbon capture, renewables.

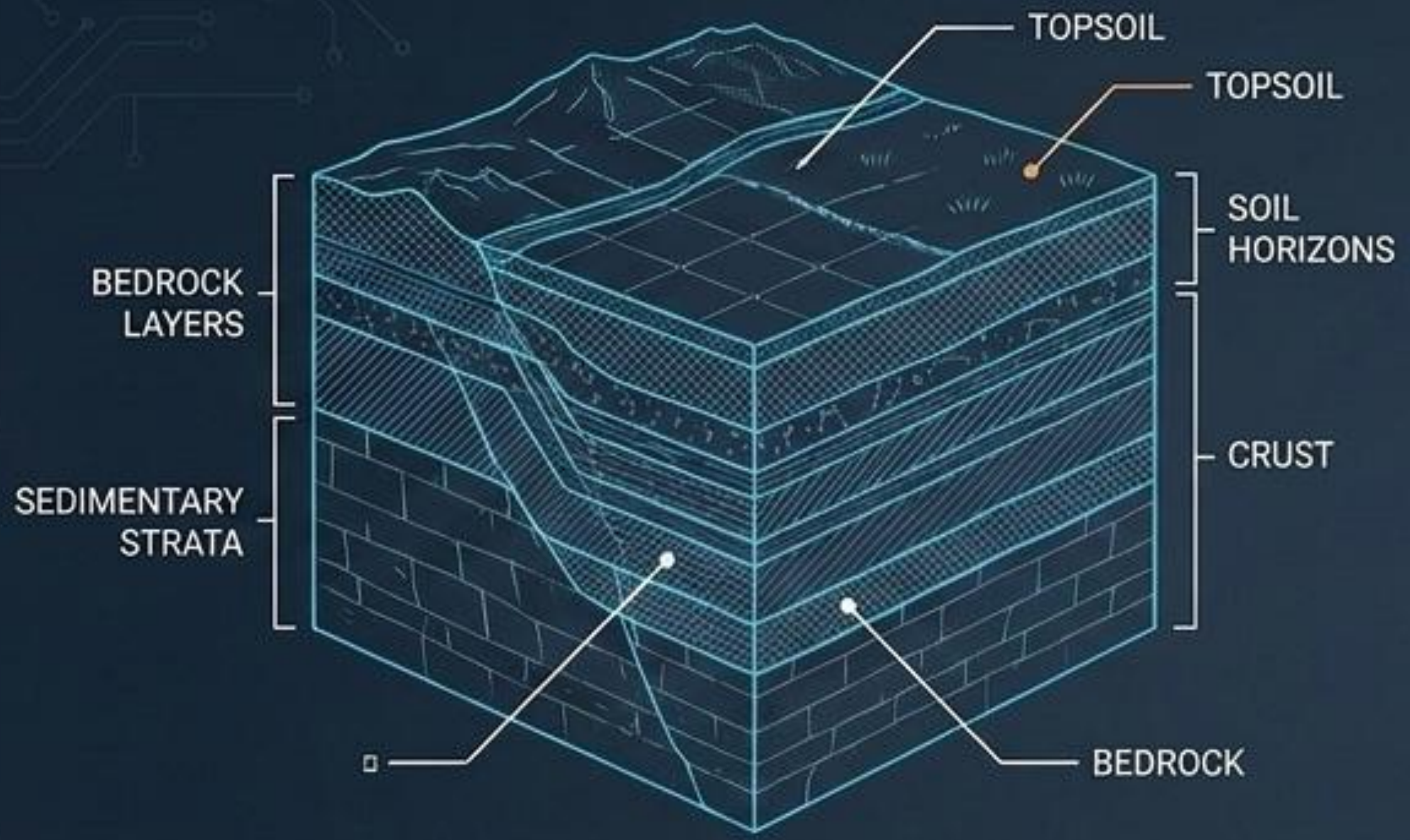
Hydrosphere



Functions: Agriculture, drinking, hydropower, climate regulation.

Stress Test (Bangladesh): A vital network of over 700 rivers facing severe pressure from pollution, flooding, and dry-season scarcity.

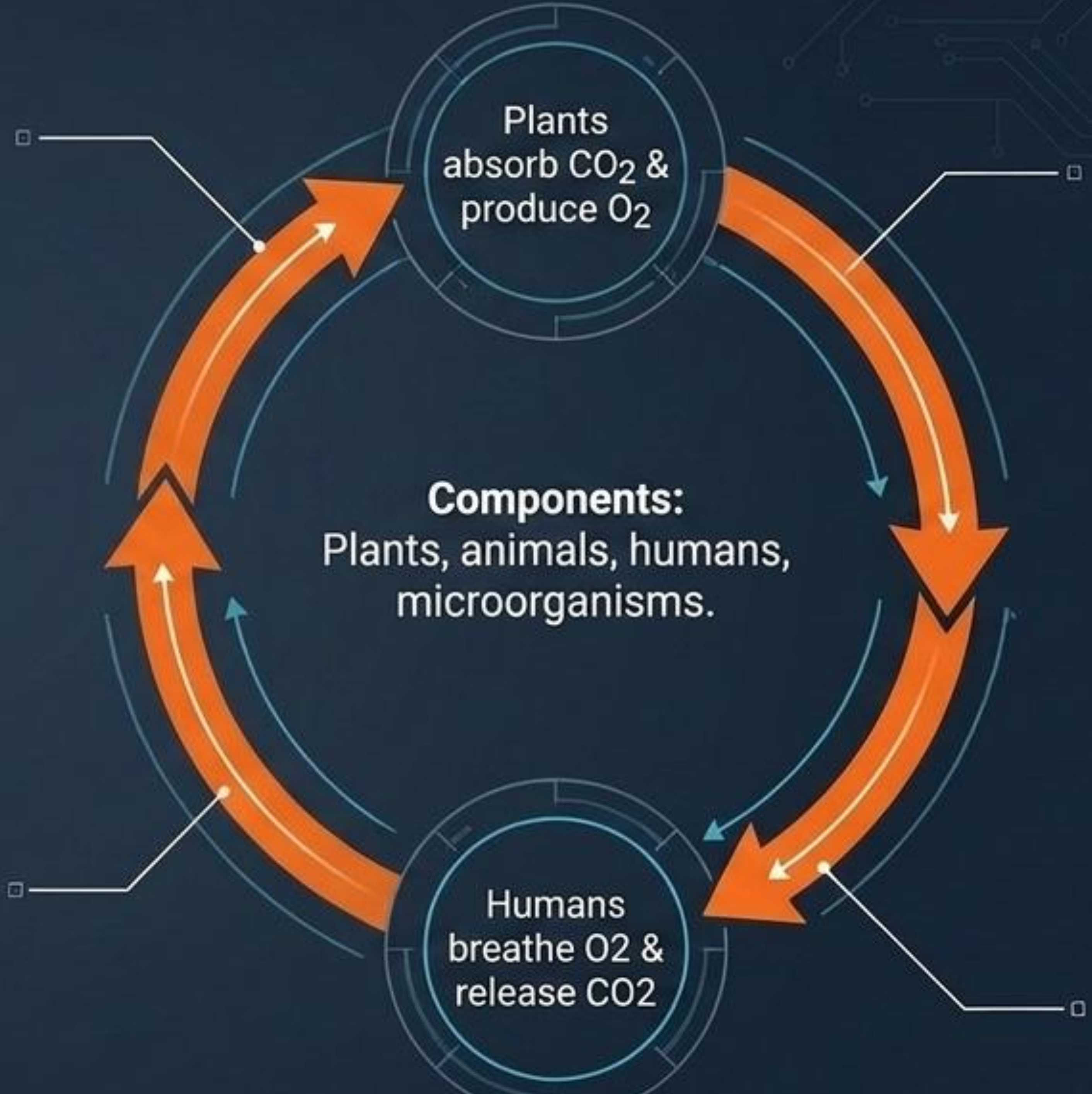
Lithosphere



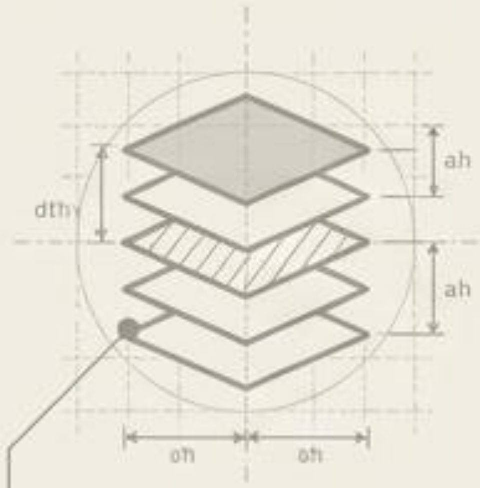
Components: Rocks, minerals, soil.
Importance: Agriculture, construction, mining, forestry.

Stress Test (Bangladesh): Severe riverbank erosion causing loss of fertile land; urban land degradation.

Biosphere



THE ATMOSPHERE AND HYDROSPHERE REGULATE GLOBAL CLIMATE BUT FACE SEVERE LOCAL STRAIN



ATMOSPHERE

COMPOSITION

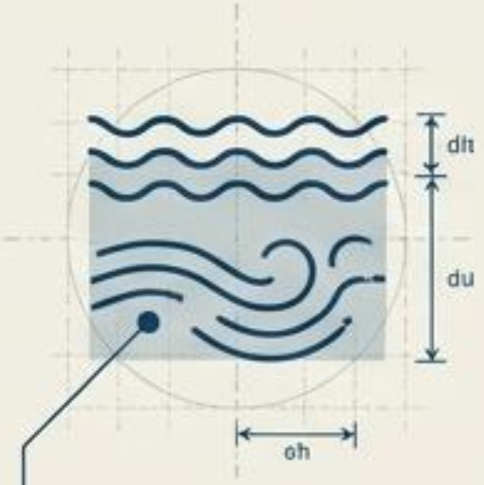
78%	NITROGEN
21%	OXYGEN
0.93%	ARGON
0.04%	CARBON DIOXIDE

FUNCTION

Provides oxygen, blocks UV radiation, controls weather.

LOCAL STRESSOR (BANGLADESH)

Dhaka frequently ranks among the worst cities globally for air quality, driven by brick kilns, trafic emissions, and construction dust.



HYDROSPHERE

COMPOSITION

Covers ~71% of Earth's surface. Crucially, only 2.5% is freshwater, and <1% is readily accessible.

FUNCTION

Drinking, agriculture, industry, climate regulation.

LOCAL STRESSOR (BANGLADESH)

Traversed by over 700 rivers, making it highly vulnerable to river pollution, wet-season flooding, and dry-season water scarcity.

THE LITHOSPHERE AND BIOSPHERE DRIVE PHYSICAL INFRASTRUCTURE AND THE CYCLE OF LIFE

LITHOSPHERE

COMPOSITION

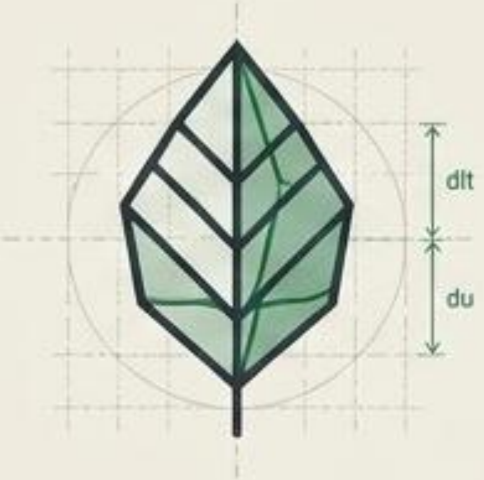
Earth's solid outer layer, housing rocks, minerals, and soil.

FUNCTION

The basis for agriculture, construction, mining, and forestry.

LOCAL STRESSOR (BANGLADESH)

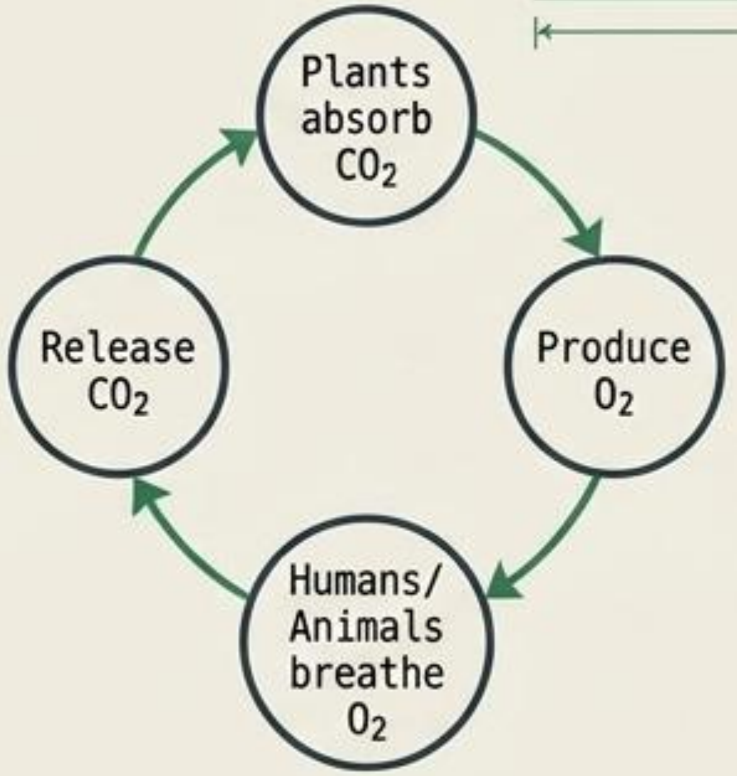
Severe riverbank erosion causing the loss of fertile agricultural land, compounded by urban land degradation.



BIOSPHERE

COMPOSITION

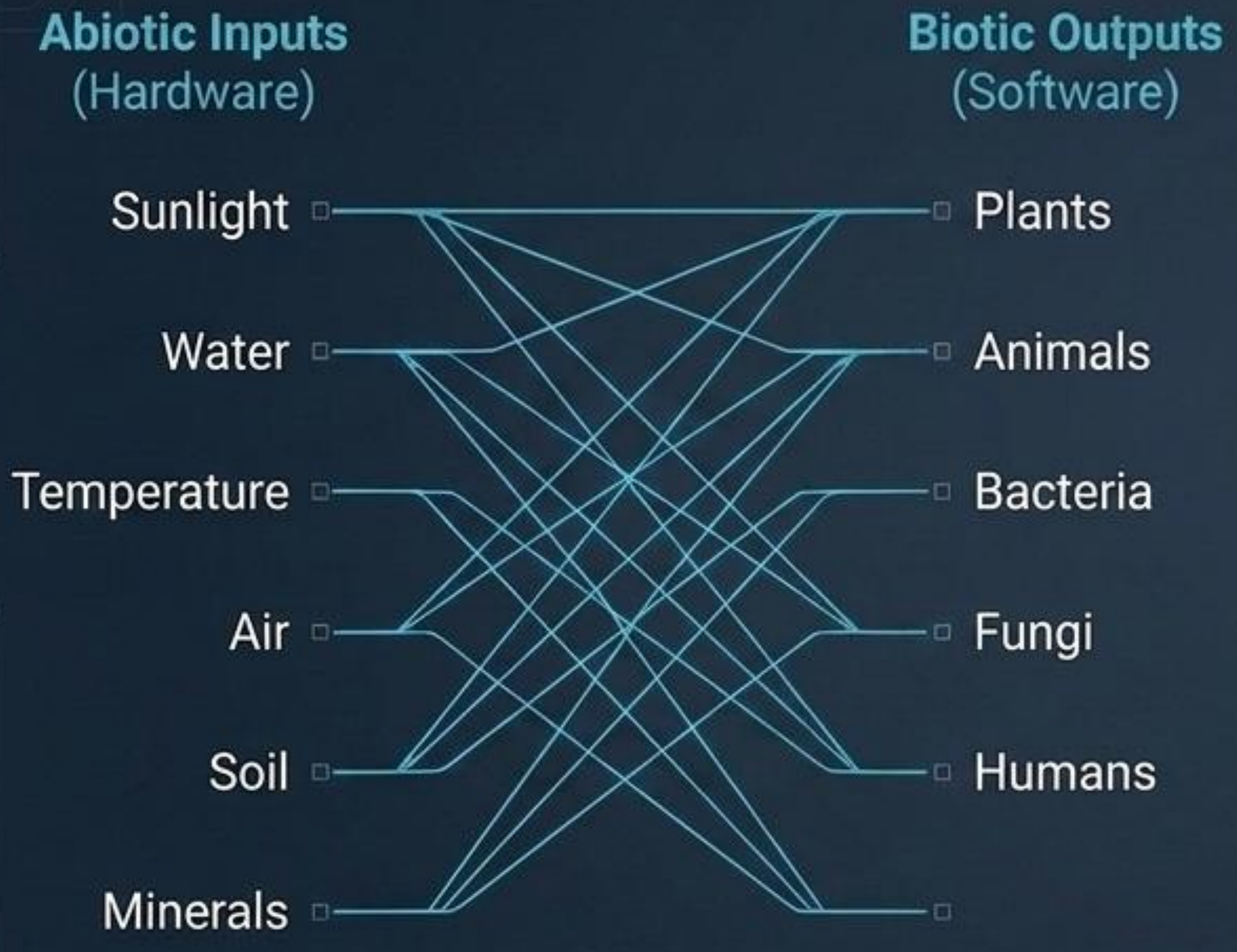
Plants, animals, humans, microorganisms.



KEY INSIGHT

The biosphere is entirely self-regulating when natural cycles remain unbroken.

Ecosystems: The Interface of Hardware and Software



ECOSYSTEMS FUNCTION AS CLOSED-LOOP ENGINES WITH ZERO WASTED OUTPUT

ECOSYSTEM DEFINITION:

The specific interaction site between living organisms and their physical environment.

BIOTIC (LIVING) INPUTS

Plants, animals, bacteria, fungi, humans.

ABIOTIC (PHYSICAL) INPUTS

Sunlight, water, temperature, air, soil, minerals.

Ecological Infrastructure of Bangladesh

System Types:

- Forest
- Grassland
- Marine
- Freshwater
- Wetlands
- Urban
- Agricultural

Haor Region: Wetland Ecosystem

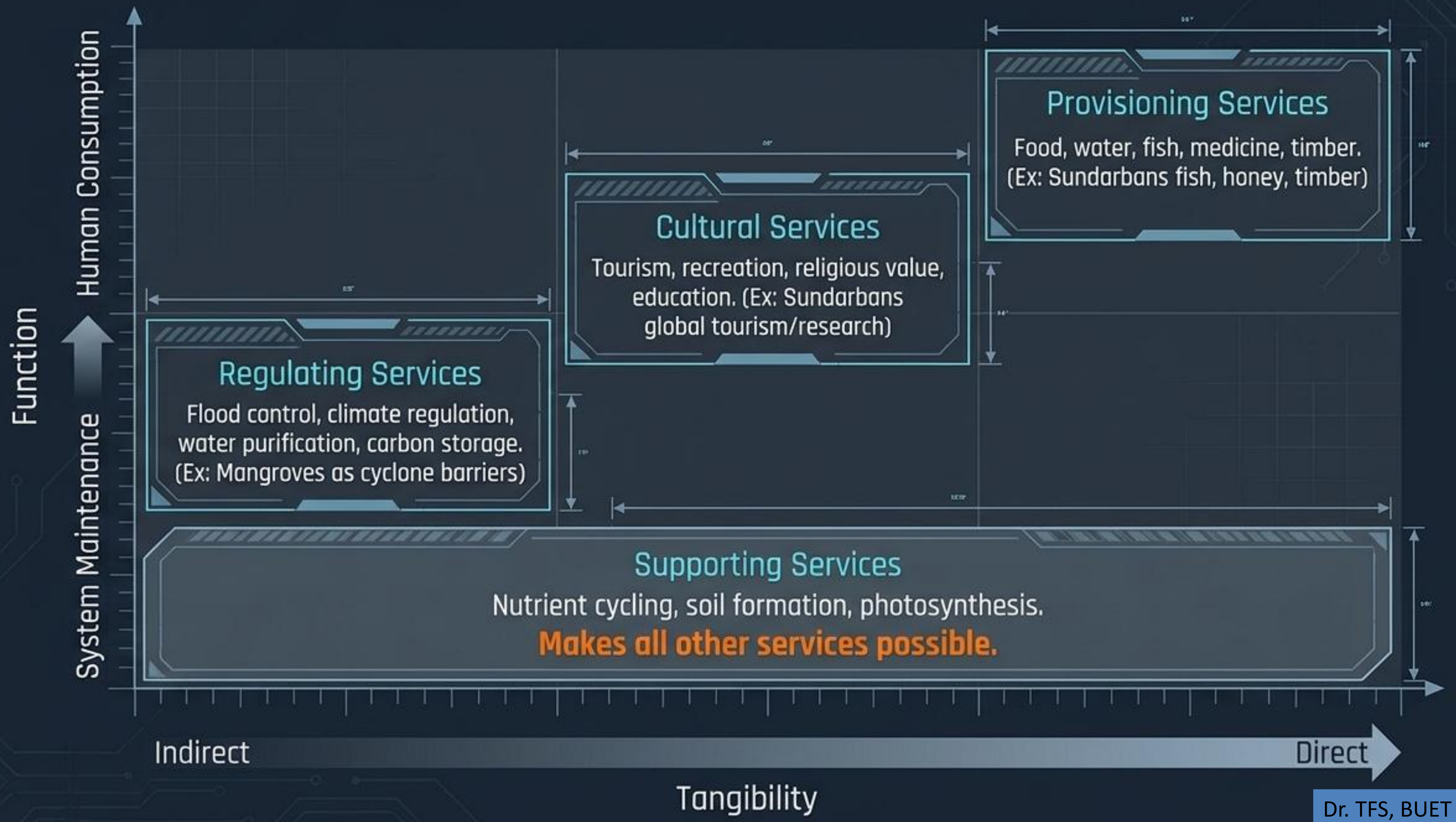
Urban Dhaka: Urban Ecosystem

Sundarbans: Mangrove Ecosystem

Cox's Bazar: Coastal Ecosystem

Saint Martin's Island: Coral Ecosystem

Ecosystem Services Diagnostic Matrix



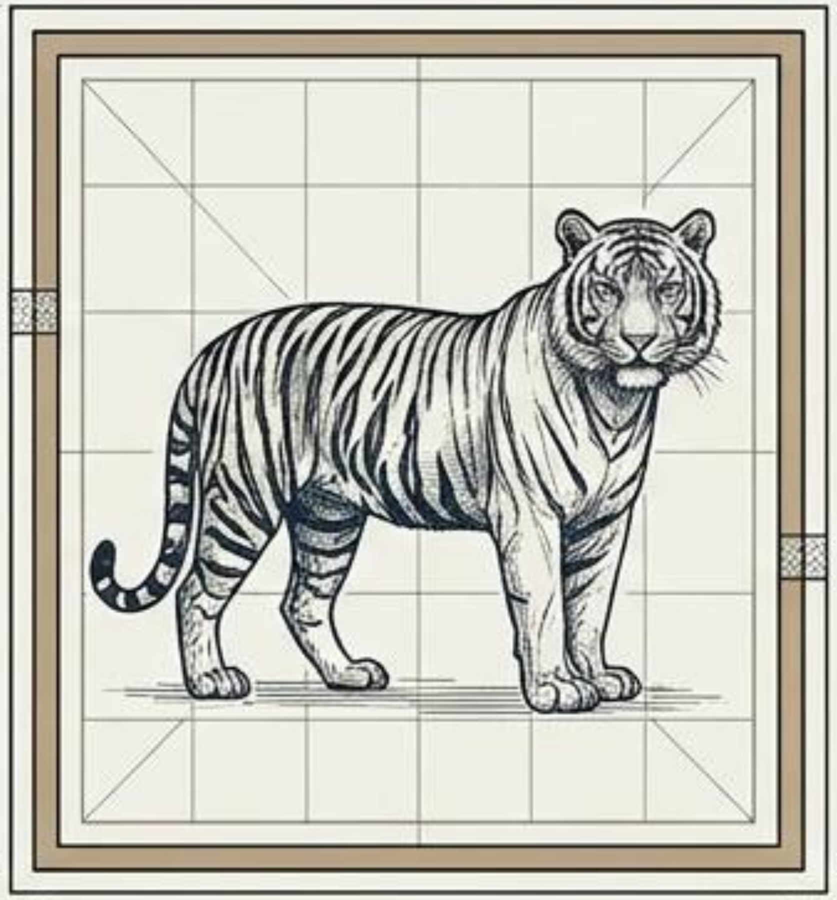
Biodiversity is the operational engine, functioning across three scales of resolution

Biodiversity: The total variety of life on Earth at all levels of organization.

Zooming Architecture

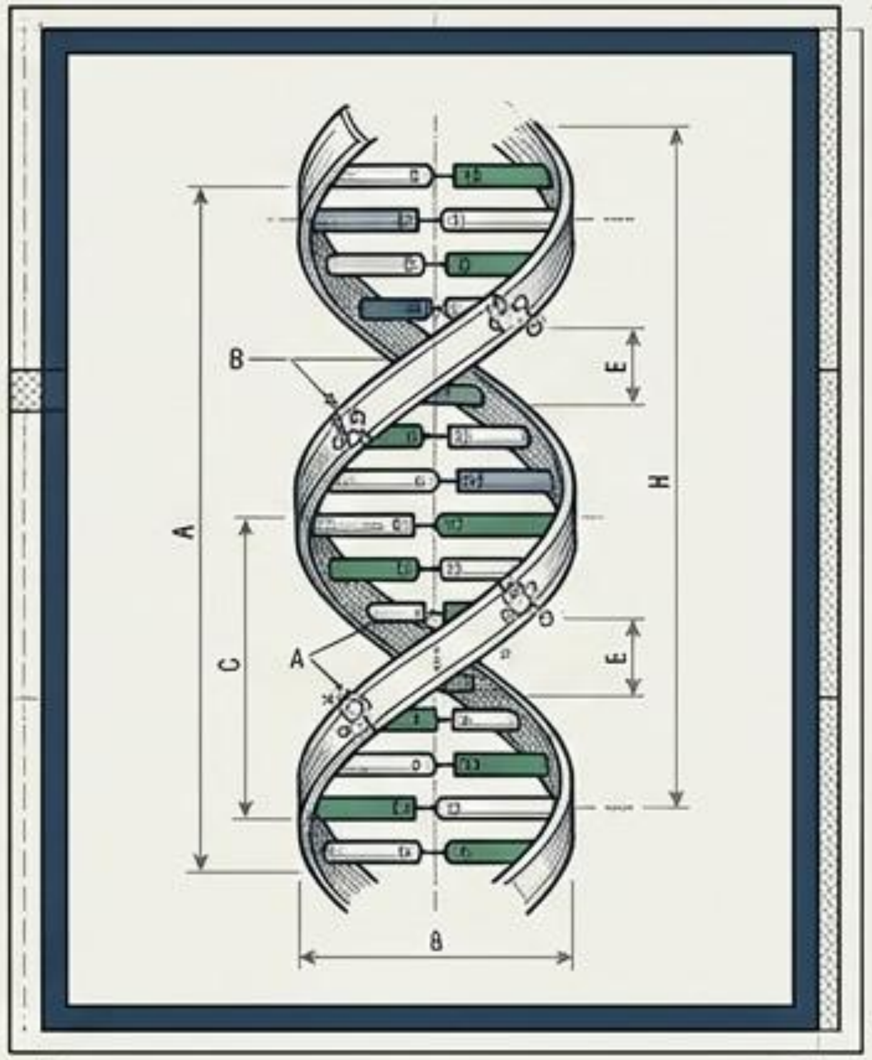
Scale 1: Ecosystem Diversity (Macro)

- Variety of habitats (e.g., Forests, wetlands, agricultural lands, coral reefs).



Scale 2: Species Diversity (Mid)

- Variety of distinct organisms.
- Bangladesh Examples: Royal Bengal Tiger, Deer, Crocodile, Kingfisher.



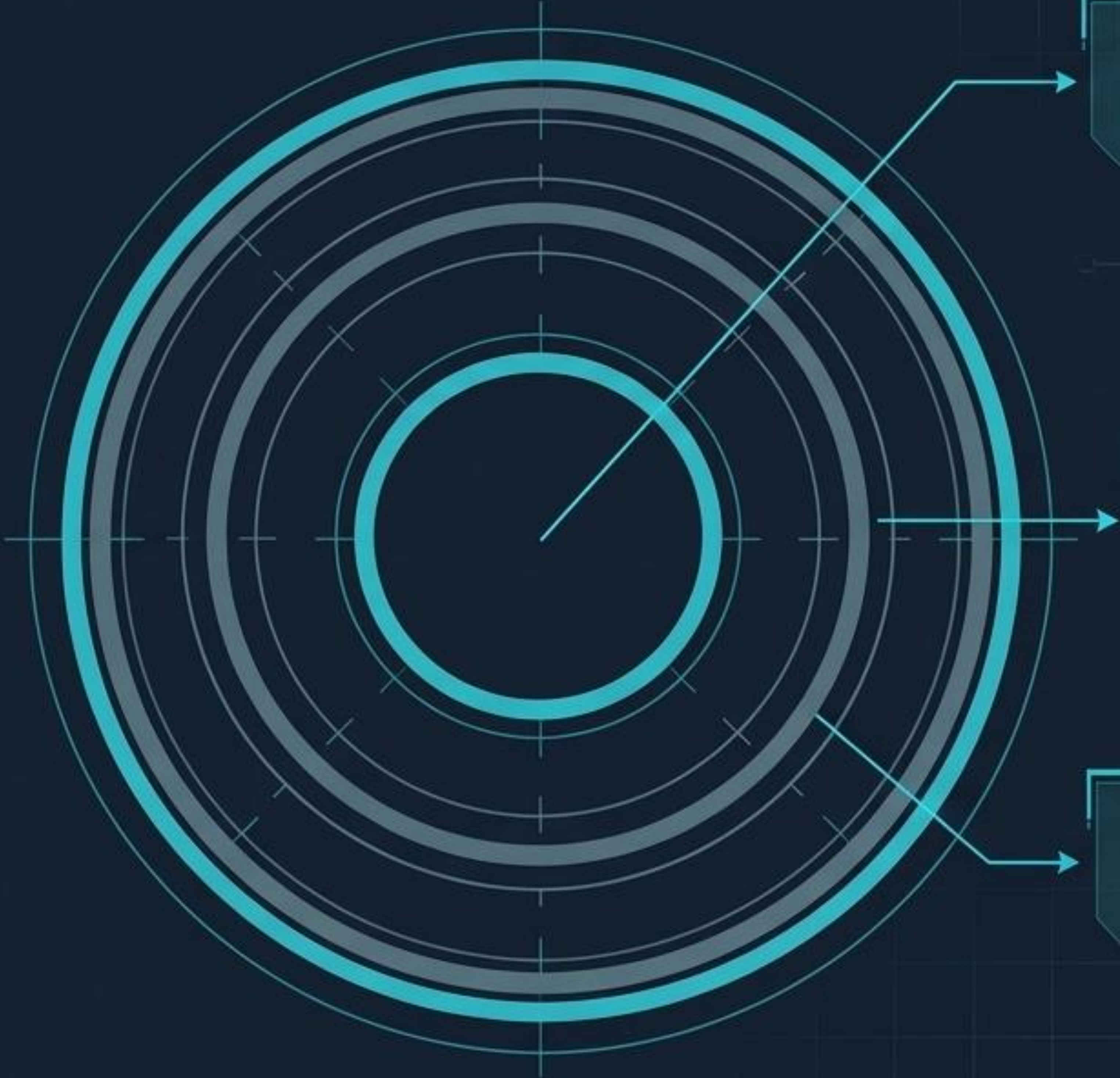
Scale 3: Genetic Diversity (Micro)

- Variation within a single species.
- Bangladesh Example: The wide variety of distinct rice strains cultivated locally.

Biodiversity: The Codebase of Resilience

Importance Vectors

- Food security
- Medicine
- Climate regulation
- Economic development
- Scientific discovery
- Disaster resilience

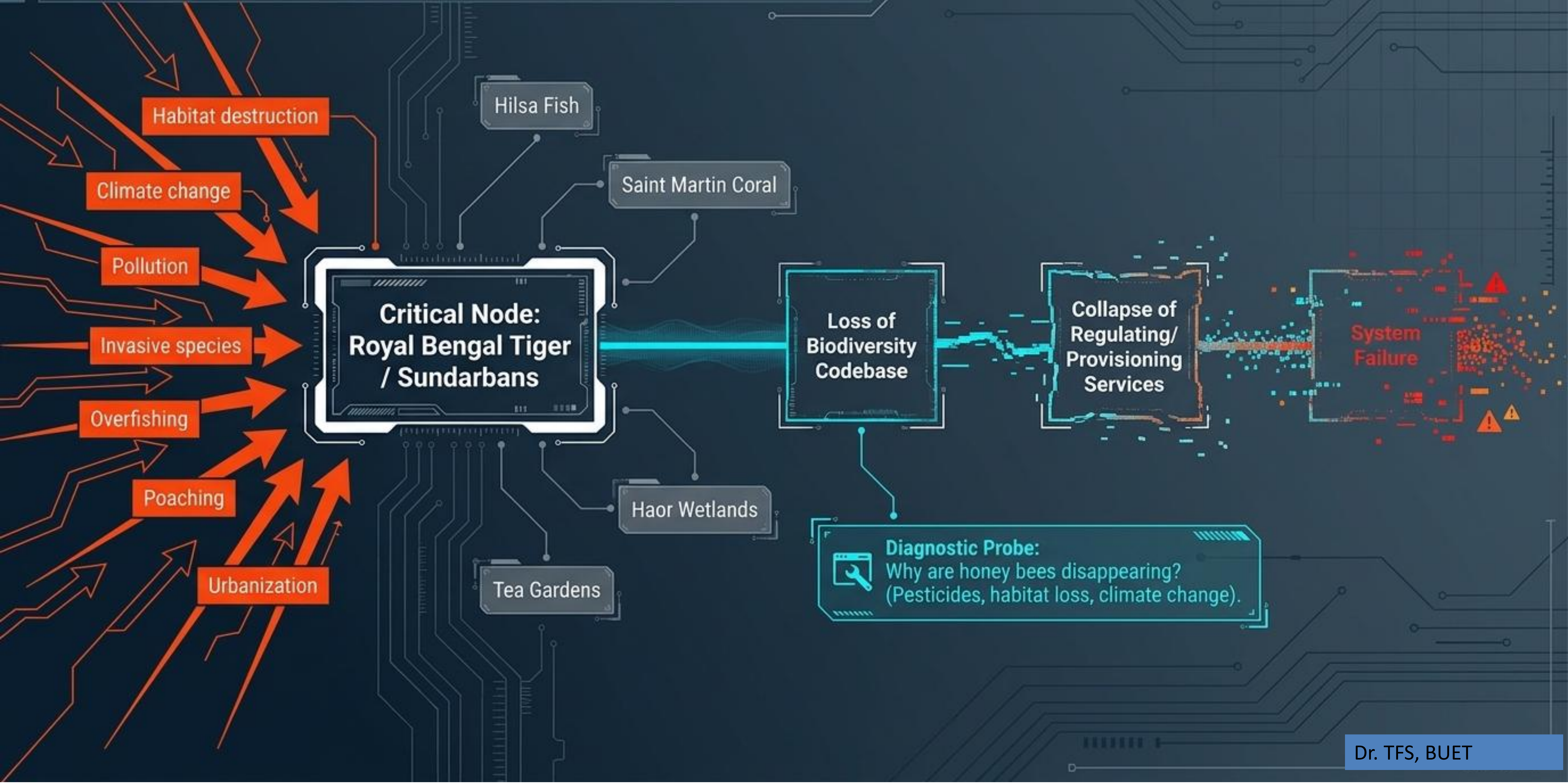


Genetic Diversity: Variation within a species. (Ex: Different rice varieties in Bangladesh).

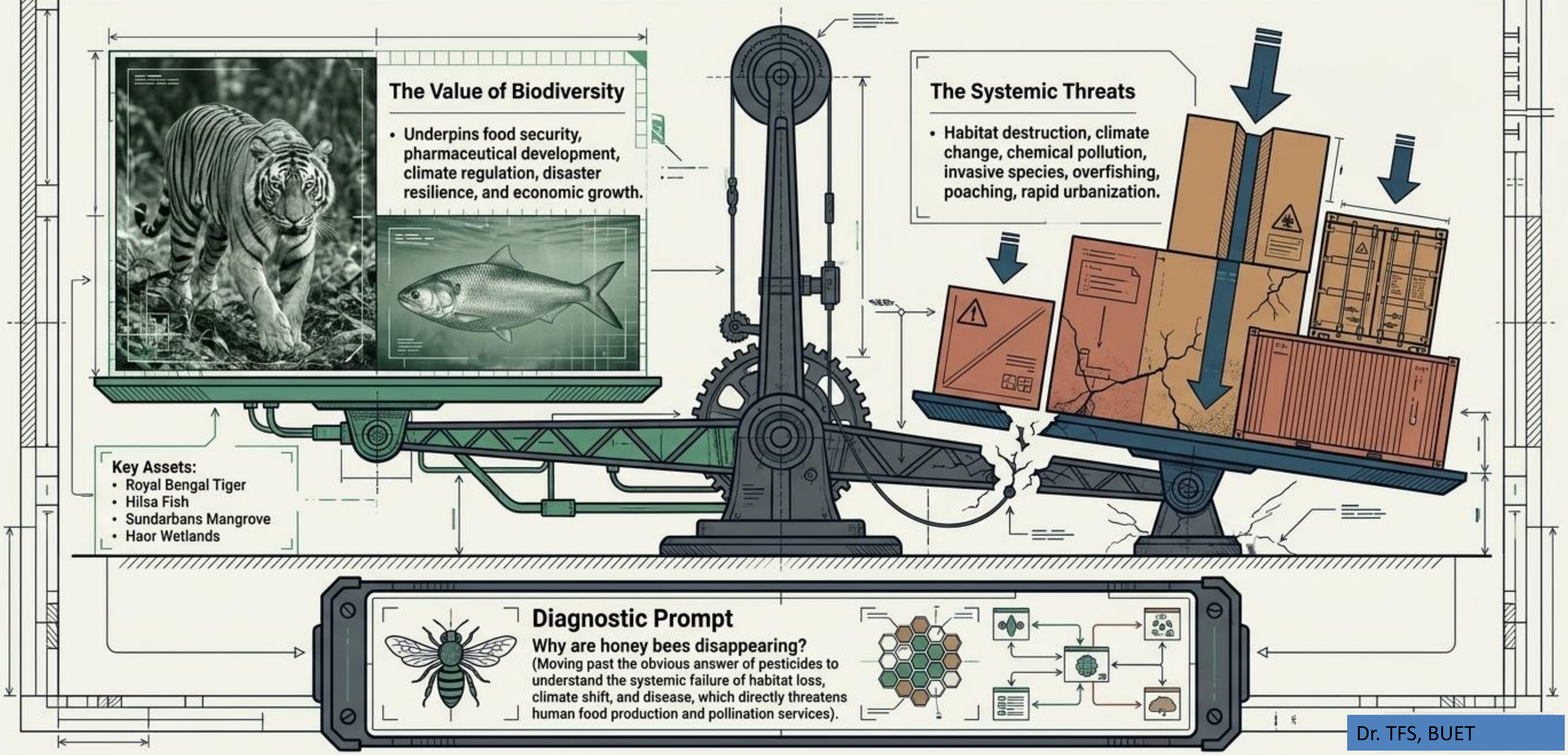
Species Diversity: Different species interacting. (Ex: Tiger, Deer, Monkey, Crocodile, Kingfisher).

Ecosystem Diversity: Different habitats operating simultaneously. (Ex: Forest, Wetland, River, Coral reef).

Systemic Threats and Cascading Failures



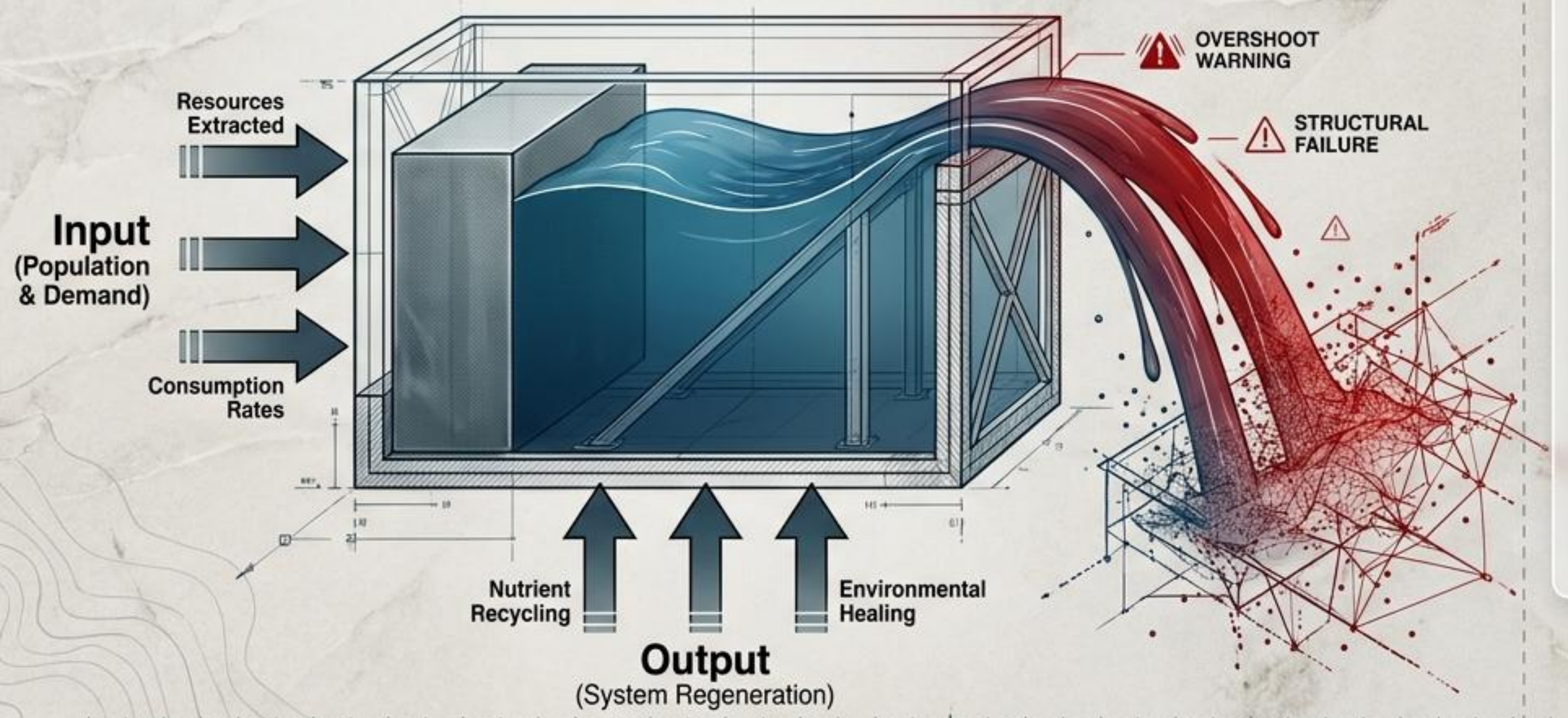
Systemic threats are eroding the biological assets critical for human survival




Engineering
Within Ecological Limits: From
Planetary Boundaries to Local Systems
Dr. TFS, BUET

Nature Operates Within Absolute Structural Limits

Carrying Capacity: The maximum population an environment can support indefinitely without degradation.



System Stress Tests:

Lake Ecosystem:

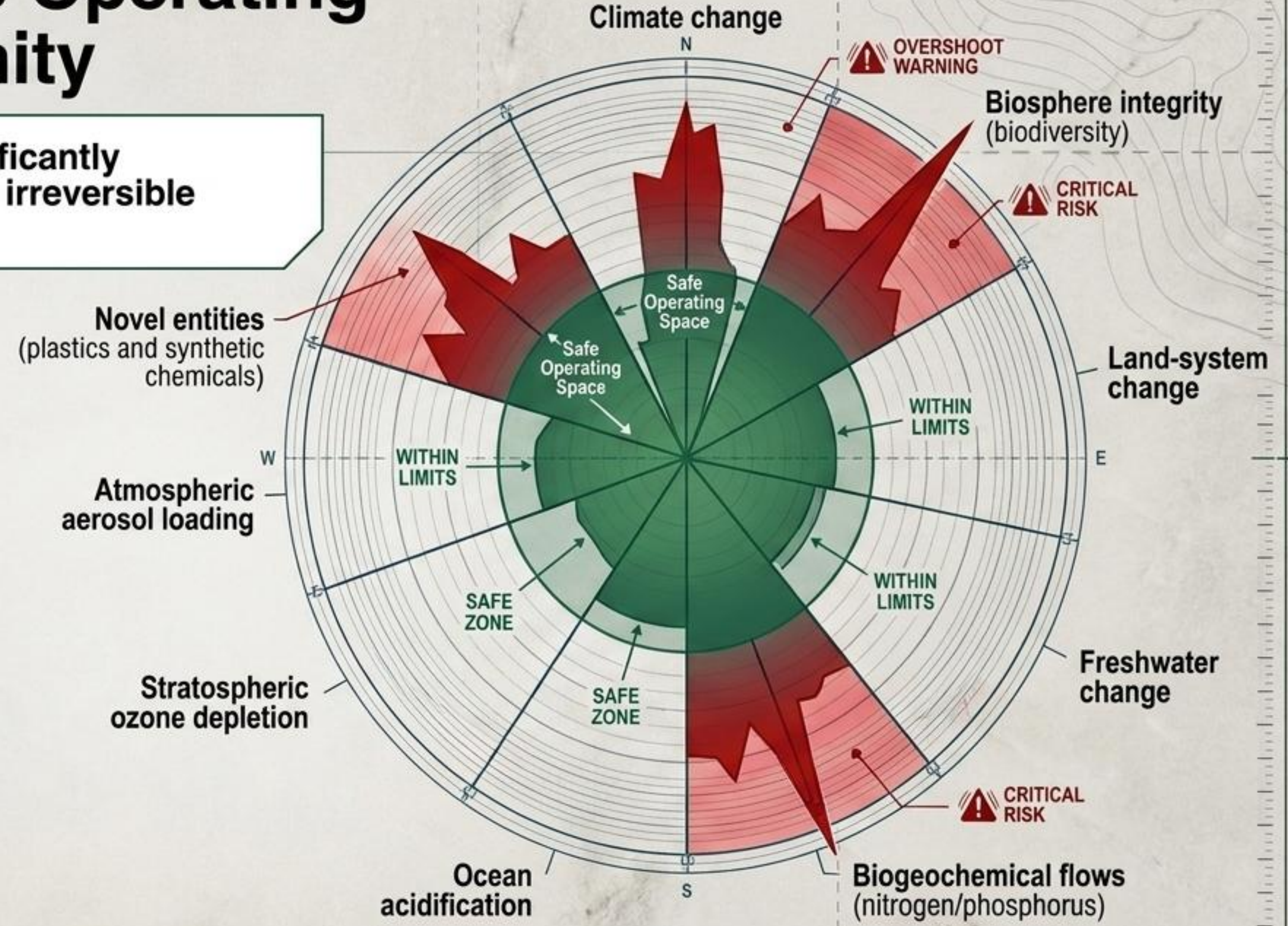
- Inputs exceed regeneration → food becomes scarce, water quality declines, fish populations collapse.

Urban Infrastructure (Dhaka):

- Rapid population growth → critical pressure on housing, transport, water supply, sanitation, and waste management.

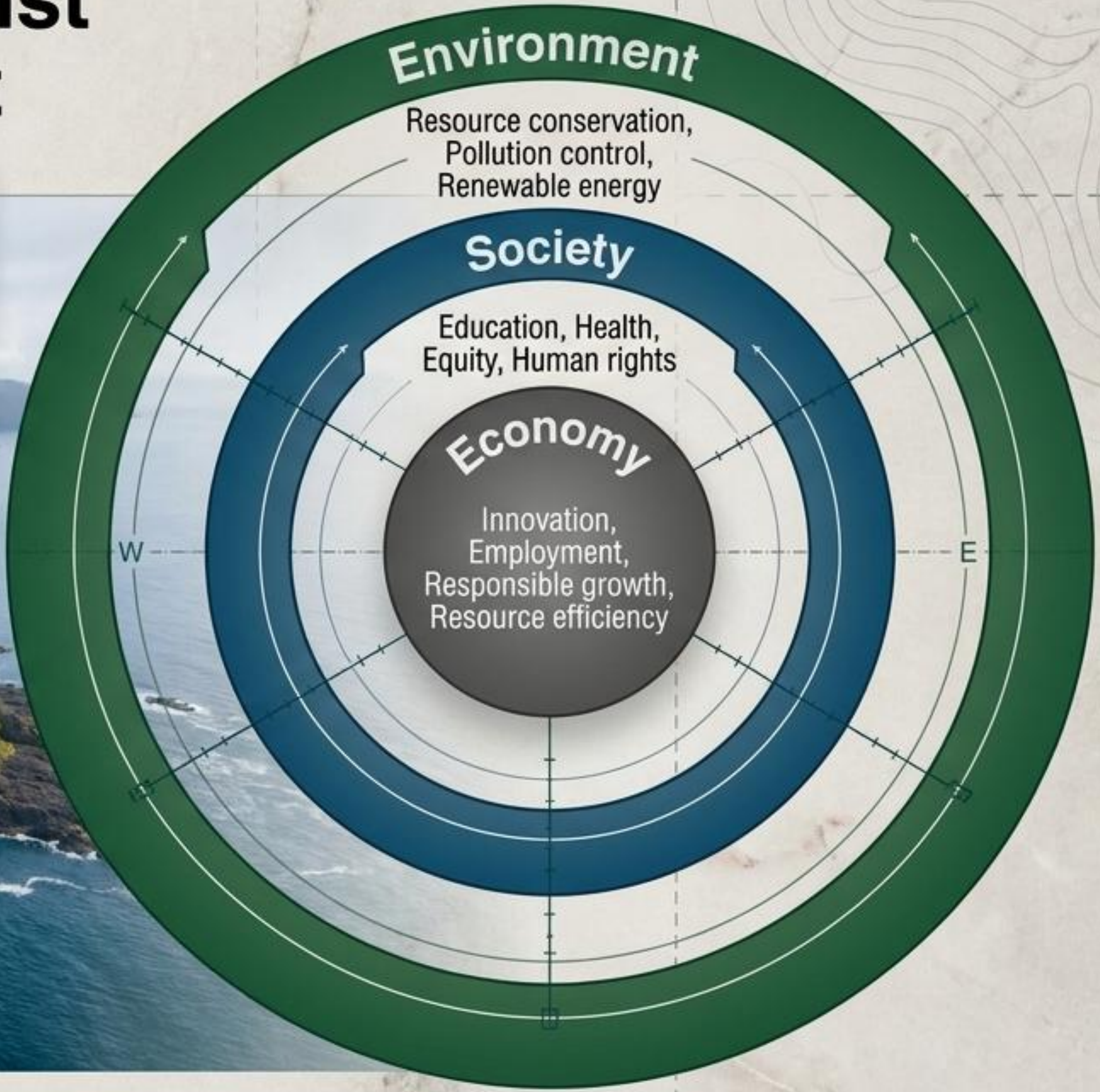
Defining the Safe Operating Space for Humanity

Crossing these boundaries significantly increases the risk of large-scale, irreversible environmental instability.



The Economy Cannot Exist Outside the Environment

Sustainability (Brundtland Commission, 1987): Development that meets the needs of the present without compromising the ability of future generations to meet their own needs.



Balancing the Pillars in Practice: Solar Home Systems

Engineering solutions must satisfy environmental, social, and economic metrics simultaneously, while anticipating long-term maintenance challenges.

Environmental Impact

Reduction in kerosene dependence and localized emissions.

Social Impact

Expanded rural electricity access, improved indoor health, extended hours for education.

Economic Impact

Localized job creation and micro-economies.

System Challenges

Maintenance, battery disposal, and long-term affordability.

The 2030 Global Engineering Agenda

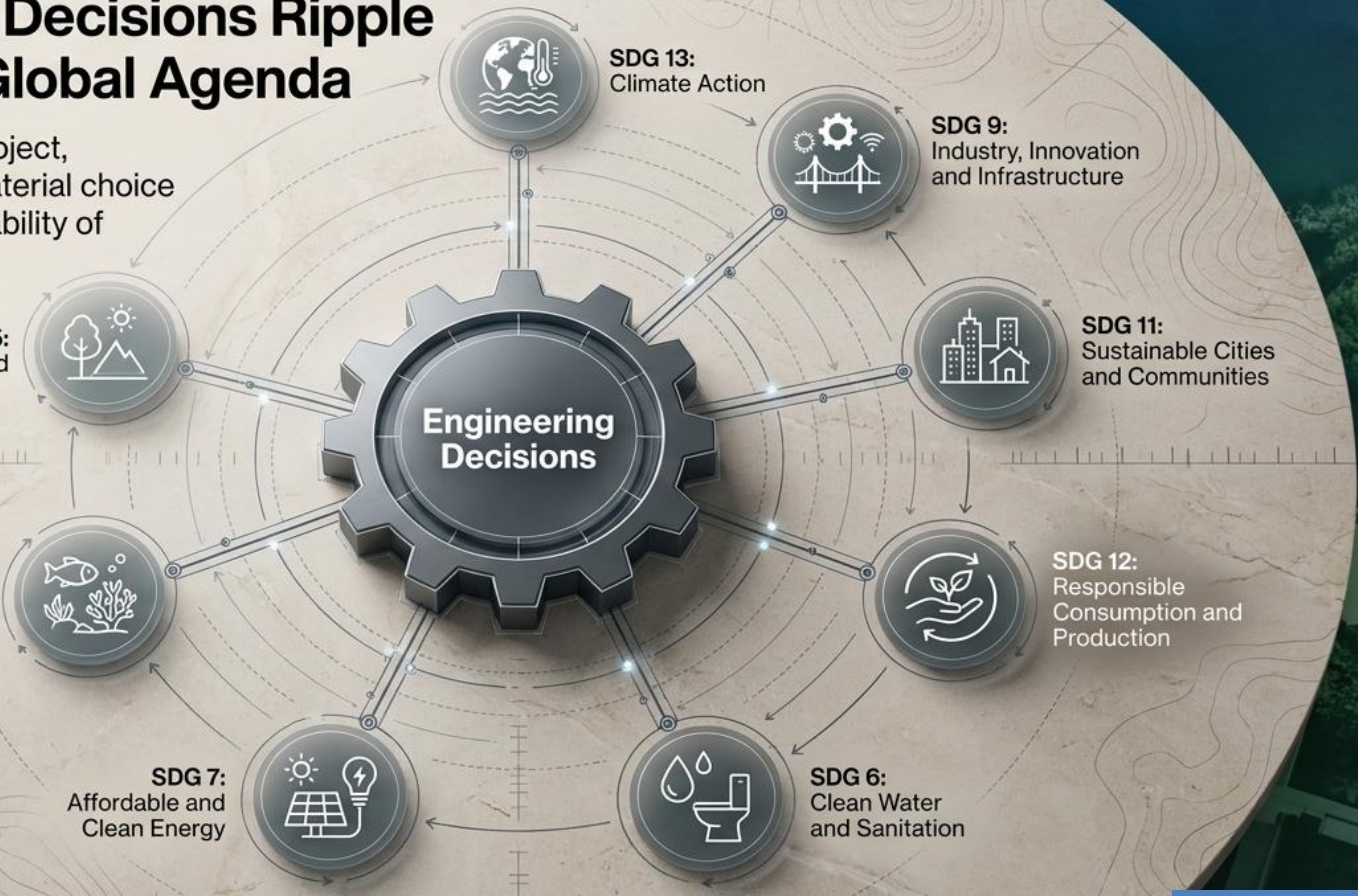
1. Adopted by the United Nations in 2015, the 17 Sustainable Development Goals (SDGs) provide a universal framework to achieve sustainable development by 2030.

2. These goals translate the concept of sustainability into measurable, actionable targets for governments, industries, and technologists worldwide.

TIME IS RUNNING OUT

Engineering Decisions Ripple Across the Global Agenda

Every infrastructure project, energy system, and material choice directly impacts the viability of the 2030 targets.



Ecosystem Diagnostic: The Sundarbans Mangrove

Core System Value

World's largest mangrove forest. Royal Bengal Tiger habitat. Critical natural barrier protecting coastal communities from cyclones. Massive carbon storage capacity. Supports regional fisheries.

Primary System Threats

Sea-level rise, creeping salinity intrusion, illegal logging, and industrial pollution.

Engineering Leverage Points

Coastal defense engineering, hydrological monitoring networks, remote sensing for conservation.

Ecosystem Diagnostic: Dhaka Urban Infrastructure

Core System Value

Economic engine of Bangladesh. High-density urban ecosystem.

Primary System Threats

Severe air pollution driven by traditional brick kilns, dense vehicle emissions, construction dust, and industrial exhaust.

System Impacts

High rates of respiratory diseases, drastically reduced visibility, and massive compounding economic costs.

Engineering Leverage Points

Implementation of cleaner kiln technologies, systemic public transport overhaul, strict emission control standards, and urban greening infrastructure.

Ecosystem Diagnostic: Saint Martin's Island

Core System Value

Bangladesh's only coral-bearing island. Hotspot for marine biodiversity. Crucial driver for the localized tourism economy.

Primary System Threats

Rampant plastic pollution, rapid coral degradation, unplanned and unregulated tourism expansion, and coastal erosion.

Engineering Leverage Points

Closed-loop waste management systems, sustainable eco-tourism infrastructure, and localized coastal erosion control mechanisms.